

Selective Inference via the Condition on Selection Framework: Inference after Variable Selection

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Slides at

http://web.stanford.edu/~jdl17/selective_inference_regression.pdf

Selective Inference

Selective Inference is about

- Testing hypotheses suggested by the data.
- Estimating quantities after seeing the data.

The quantities of interest are only specified after seeing the data!

Selective Inference is common. In many applications there is no hypothesis specified before data collection.

- Inference after variable selection.
- Exploratory Data Analysis by Tukey emphasized using data to suggest hypotheses, and post-hoc analysis to test these.
- Screening in Genomics, only select those with large t -statistic.
- Peak/bump hunting in neuroscience, only study process when $X_t > \tau$ and critical points of the process.

Conventional Wisdom (Data Dredging, Wikipedia)

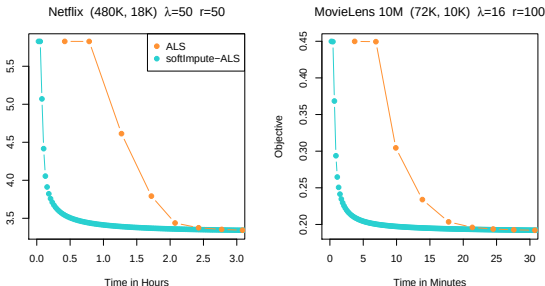
A key point in proper statistical analysis is to test a hypothesis with data that was not used in constructing the hypothesis. (Data splitting)

This talk

The Condition on Selection framework allows you to specify and test hypotheses using the same dataset.

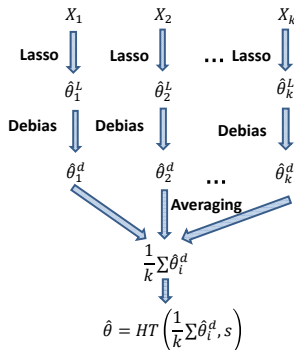
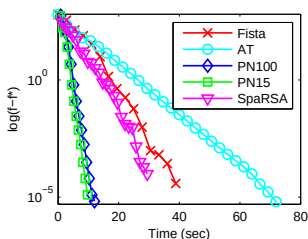
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Scalable and Distributed Algorithms I



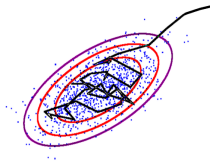
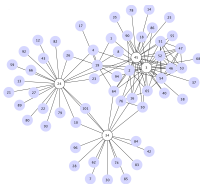
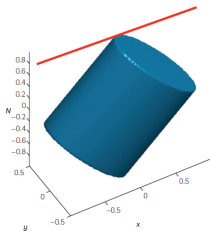
- 1 **Matrix Factorization.** fast-ALS reduces the computational cost and communication of ALS by $O(r)$, but with the same convergence rate. Apache Spark implementation takes 6 seconds per iteration for a $10^6 \times 10^6$ matrix with 10^9 non-zeroes, and 2 minutes per iteration for a $10^7 \times 10^7$ matrix with 10^9 non-zeroes.
- 2 **One-Pass Non-negative Matrix Factorization on Hadoop** Factorizes a 2TB heat transfer dataset in 50 min and factorizes a 345GB flow cytometry dataset in 20 min.

Scalable and Distributed Algorithms II



- 1 **Proximal Newton Method.** Designed and analyzed a second order algorithm for regularized estimators, which includes state-of-art methods such as `glmnet`, `liblinear`, and `quic`.
- 2 **Communication-efficient distributed sparse regression.** Proposed an algorithm for solving distributed sparse regression, which has the lowest communication cost among algorithms that achieve estimation rate $\left\| \hat{\beta} - \beta^0 \right\|_2 \leq \sqrt{\frac{s \log p}{n}}$.

High-dimensional model selection and estimation



- **Unified theory of model selection consistency in regularized estimators** of the form $\ell(\theta) + \rho(\theta)$. Provides structure learning consistency in graphical models, variable selection consistency in lasso, and rank consistency for nuclear norm (matrix completion).
- **Graphical model structure learning with mixed data.**
- **Confidence intervals via stochastic gradient sampling.** SGD is one of the most common learning algorithms. We use iterates of SGD to build confidence intervals in the spirit of bootstrap and MCMC.

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Motivation: Linear regression in high dimensions

- 1 Select relevant variables $\hat{M}$ via a variable selection procedure (k most correlated, lasso, forward stepwise ...).
- 2 Fit linear model using only variables in $\hat{M}$, $\hat{\beta}^{\hat{M}} = X_{\hat{M}}^{\dagger} y$.
- 3 Construct 90% z-intervals $(\hat{\beta}_j - 1.65\sigma_j, \hat{\beta}_j + 1.65\sigma_j)$.

Are these confidence intervals correct?

- Generate design matrix $X \in \mathbf{R}^{n \times p}$ from a standard normal with $n = 20$ and $p = 200$.
- Let $y = \mathcal{N}(X\beta^0, 1)$.
- β^0 is 2 sparse with $\beta_1^0, \beta_2^0 = SNR$.
- Use marginal screening to select $k = 2$ variables, and then fit linear regression over the selected variables.
- Construct 90% confidence intervals for β and check the coverage proportion.

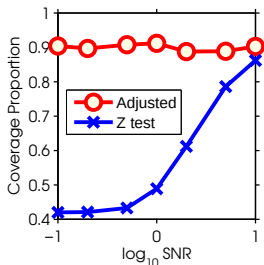


Figure: Plot of the coverage proportion across a range of SNR. The coverage proportion of the z intervals ($\hat{\beta} \pm 1.65\sigma$) is far below the nominal level of $1 - \alpha = .9$, even at $\text{SNR} = 5$. The selective intervals (our method) always have coverage proportion .9.

Remember this simulation.

We will return to explain the results of the simulation.

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Notation

- The selection function $\hat{H}$ selects the hypothesis of interest, $\hat{H}(y) : \mathcal{Y} \rightarrow \mathcal{H}$.
- $\phi(y; H)$ be a test of hypothesis H , so reject if $\phi(y; H) = 1$.
- $\phi(y; H)$ is a valid test of H if $\mathbb{P}_0(\phi(y; H) = 1) \leq \alpha$.
- $\{y : \hat{H}(y) = H\}$ is the selection event.
- $F \in N(H)$ if F is a null distribution with respect to H .

Definition

$\phi(y; \hat{H})$ is a valid selective test if

$$\mathbb{P}_F(\phi(y; \hat{H}(y)) = 1 | F \in N(\hat{H})) \leq \alpha$$

- Reduction to Simultaneous Inference: Assume that there is an apriori set of hypotheses $\mathcal{H}$ that could be tested. We can simultaneously control the type 1 error over all of $\mathcal{H}$, which implies selective type 1 error rate control for some selected $\hat{H}(y) \in \mathcal{H}$. Scheffe's method and POSI (Berk et al. 2013).
- Data Splitting: Split the dataset $y = (y_1, y_2)$. Let $\hat{H}(y_1)$ be the selected hypothesis, and construct the test of $\hat{H}(y_1)$ using only y_2 .

Conditioning for Selective Type 1 Error Control

We can design a valid selective test ϕ by ensuring ϕ is a **valid test with respect to the distribution conditioned on the selection event meaning**

$$\forall F \in N(H_i), \mathbb{P}_F(\phi(y; H_i) = 1 | \hat{H} = H_i) \leq \alpha,$$

then

$$\begin{aligned} & \mathbb{P}_F(\phi(y; \hat{H}(y)) = 1 | F \in N(\hat{H})) \\ &= \sum_{i: F \in N(H_i)} \mathbb{P}_F(\phi(y; H_i) = 1 | \hat{H} = H_i) \mathbb{P}_F(\hat{H} = H_i | F \in N(\hat{H})) \\ &\leq \alpha \sum_{i: F \in N(H_i)} \mathbb{P}_F(\hat{H} = H_i | F \in N(\hat{H})) \end{aligned}$$

We have reduced the problem to designing a conditional test

$$\mathbb{P}_F(\phi(y; H_i) = 1 | \hat{H} = H_i) \leq \alpha.$$

Example: Maximum of a Normal

Let $y \sim \mathcal{N}(\mu, \Sigma)$. We would like to test $H_i : \mu_i = 0$ against the alternative $\mu_i > 0$. Let $\hat{H}(y) = H_{i^*}$, where y_{i^*} is the maximum *i.e.* $y_{i^*} = \max_i y_i$. We can make a test by rejecting when $y_{i^*} > c$. Condition on selection tells us to choose the cutoff c so type 1 error rate under $y_{i^*} | \{y_{i^*} > \max_{i \neq i^*} y_i\}$ is α .

Outline of the Rest of the Talk

- 1 Review linear regression, and cast inference after variable selection as a selective inference problem.
- 2 Many variable selection procedures have selection events $\{y : \hat{M}(y) = M\} = \{y : Ay \leq b\}$, where M is the subset of variables selected.
- 3 An exact, valid selective test ϕ can be constructed for linear functions $\eta(\hat{M}(y))^T \mu$, using the Condition on Selection framework without sampling or numerical integration. This is done by deriving the conditional distribution of $\eta^T y$, which will be a truncated normal.
- 4 A selective test ϕ can be inverted to make a selective confidence interval. These confidence intervals control false coverage rate, a selective type 1 error metric (Benjamini & Yekutieli 2005).

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Model

- Assume that $y_i = \mu(x_i) + \epsilon_i$
- $\epsilon_i \sim \mathcal{N}(0, \sigma^2)$.
- $x_i \in \mathbf{R}^p$, $y \in \mathbf{R}^n$, and $\mu = \begin{bmatrix} \mu(x_1) \\ \vdots \\ \mu(x_n) \end{bmatrix}$.
- Design matrix $X = \begin{bmatrix} x_1^T \\ \vdots \\ x_n^T \end{bmatrix} \in \mathbf{R}^{n \times p}$.

Review of Linear Regression

The best linear approximation is $\beta^0 = X^\dagger \mu$. Linear regression estimates this using

$$\hat{\beta} = X^\dagger y.$$

Theorem

The least squares estimator is distributed

$$\hat{\beta} \sim \mathcal{N}(\beta^0, \sigma^2 (X^T X)^{-1})$$

and

$$\Pr \left(\beta_j^0 \in \left[\hat{\beta}_j \pm z_\alpha \sigma (X^T X)^{-1/2}_{jj} \right] \right) = 1 - \alpha.$$

Explaining the simulation

- 1 The confidence intervals rely on the result that $\hat{\beta}$ is Gaussian.
- 2 The variable selection procedure (marginal screening) returns a set of variables $\hat{M}(y)$. In particular,

$$|X_{\hat{M}}^T y| > |X_{-\hat{M}}^T y|.$$

- 3 For any fixed set M , $X_M^T y$ is Gaussian, but $X_{\hat{M}}^T y$ is not Gaussian!

Example

Let $y \sim \mathcal{N}(0, I)$, and $X = I$. Let $i^* = \arg \max y_i$, then y_{i^*} is not Gaussian. The interval $y_{i^*} \pm 2$ is not a valid 95% confidence interval!

Which population parameter or What is the target?

Population parameters

- 1 **Sub-model parameter.** $\beta^M = X_M^\dagger \mu$ (advocated by the POSI group).
- 2 **OLS parameter.** In the $n \geq p$ regime without the linear model assumption, $\beta^* = X^\dagger \mu$ is the best linear approximation. Under the linear model assumption $\mu = X\beta^0$ and $n > p$, $\beta^* = \beta^0$.
- 3 **The “true” parameter for $p > n$.** Assuming a sparse linear model $\mu = X\beta^0$, the parameter of interest is β^0 .

For the talk, I will focus on inference for the sub-model parameter β^M , but analagous results hold for the other two.

Selective Inference in Linear Regression

Selective Inference reduces to testing $\eta(\hat{M}(y))^T \mu$.

- 1 Sub-model parameter. $\beta_j^{\hat{M}} = e_j^T X_{\hat{M}}^\dagger \mu = \eta(\hat{M}(y))^T \mu$, where $\eta(\hat{M}(y))^T$ is row of $X_{\hat{M}}^\dagger$.
- 2 OLS parameter. $e_j^T \beta_{\hat{M}}^* = e_j^T X^\dagger \mu = \eta(\hat{M}(y))^T \mu$.
- 3 True parameter. Unfortunately, β^0 is not necessarily a linear function of μ . Luckily under the scaling $n \gg s^2 \log^2 p$ and restricted eigenvalue assumptions, there is a parameter β^d that satisfies $\sqrt{n} \left\| \beta^d(\hat{M}) - \beta^0 \right\|_\infty = o(1)$, and β^d is a linear function of μ .

Testing regression coefficients reduce to testing an adaptive/selected linear function of μ

$$H_0 : \eta(\hat{M}(y))^T \mu = \beta.$$

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- POSI (Berk et al. 2013) widen intervals to simultaneously cover all coefficients of *all* possible submodels. POSI is an example of reducing selective to simultaneous inference. POSI is extremely conservative and is only computationally feasible for $p \leq 20$.
- Asymptotic normality by “inverting” KKT conditions (Zhang 2012, Buhlmann 2012, Van de Geer 2013, Javanmard 2013). Asymptotic result that requires consistency of the lasso, and computational cost equivalent to solving p lasso’s.
- Significance testing for Lasso (Lockhart et al. 2013) tests for whether all signal variables are found. Our framework allows us to test the same thing with no assumptions on X and is completely non-asymptotic and exact.
- Knockoff for FDR control in linear regression (Foygel and Candès 2014) allows for exact FDR control for $n \geq 2p$ without any assumptions on X .

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Algorithm 1 Marginal screening algorithm

- 1: **Input:** Design matrix X , response y , and model size k .
 - 2: Compute $|X^T y|$.
 - 3: Let $\hat{M}$ be the index of the k largest entries of $|X^T y|$.
 - 4: Compute $\hat{\beta}_{\hat{M}} = (X_{\hat{M}}^T X_{\hat{M}})^{-1} X_{\hat{M}}^T y$
-

Marginal Screening Selection Event

The marginal screening selection event is a subset of $\mathbf{R}^n$:

$$\begin{aligned} & \left\{ y : \hat{s}_i x_i^T y > \pm x_j^T y, \text{ for each } i \in \hat{M} \text{ and } j \in \hat{M}^c \right\} \\ &= \left\{ y : A(\hat{M}, \hat{s})y \leq b(\hat{M}, \hat{s}) \right\} \end{aligned}$$

The marginal screening selection event corresponds to selecting a set of variables $\hat{M}$, and those variables having signs

$$\hat{s} = \text{sign} \left(X_{\hat{M}}^T y \right).$$

Lasso Selection Event

$$\hat{\beta} = \arg \min_{\beta} \frac{1}{2} \|y - X\beta\|^2 + \lambda \|\beta\|_1$$

From KKT conditions, a set of variables $\hat{M}$ is selected with $\text{sign}(\hat{\beta}_{\hat{M}}) = \hat{s}$ iff

$$\left\{ y : \text{sign}(\beta(\hat{M}, \hat{s})) = \hat{s}, \left\| Z(\hat{M}, \hat{s}) \right\|_{\infty} < 1 \right\} = \{y : Ay \leq b\}$$

This says that the inactive subgradients are strictly dual feasible, and the signs of the active subgradient agrees with the sign of the lasso estimate.

Selection events correspond to affine regions.

$$\{\hat{M}(y) = M\} = \{Ay \leq b\} \text{ and } y | \hat{M}(y) = M \stackrel{d}{=} \mathcal{N}(\mu, \Sigma) | \{Ay \leq b\}$$

Selection event

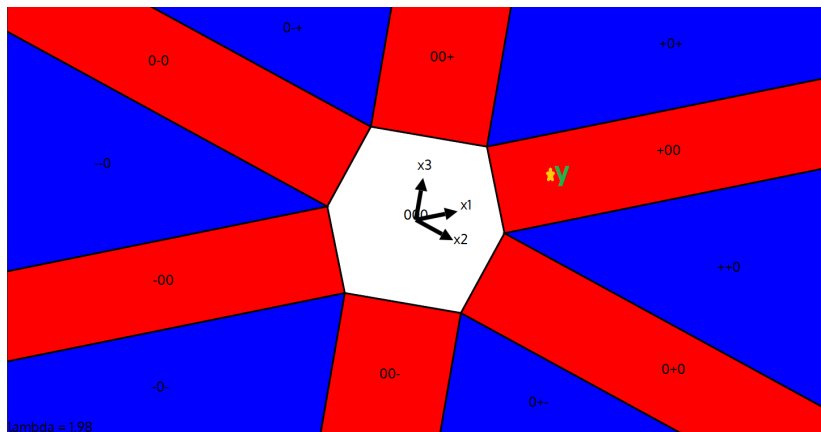


Figure: $(n, p) = (2, 3)$. White, red, and blue shaded regions correspond to different selection events. The shaded region that y falls into is where lasso selects variable 1 with positive sign.

<http://naftaliharris.com/blog/lasso-polytope-geometry/> has a javascript app that allows for visualizing the lasso selection regions.

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Constrained Gaussians

- The distribution of $y \sim \mathcal{N}(\mu, \sigma^2 I)$ conditional on $\{y : Ay \leq b\}$ has density $\frac{1}{\Pr(Ay \leq b)} \phi(y; \mu, \Sigma) \mathbb{1}(Ay \leq b)$.
- Ideally, we would like to sample from the density to approximate the sampling distribution of our statistic under the null. This is computationally expensive.
- For testing regression coefficients, we only need distribution of $\eta^T y | \{Ay \leq b\}$.

Computationally Tractable Inference

It turns out

$$\eta^T y | \{Ay \leq b, P_{\eta^\perp} y\} \stackrel{d}{=} \text{TruncatedNormal}.$$

Using this distributional result, we avoid sampling and integration.

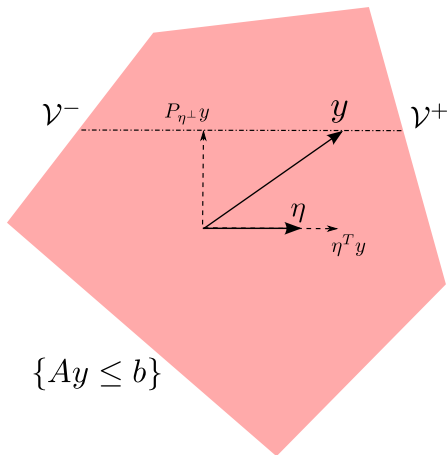


Figure: A picture demonstrating that the set $\{Ay \leq b\}$ can be characterized by $\{\mathcal{V}^- \leq \eta^T y \leq \mathcal{V}^+\}$. Assuming $\Sigma = I$ and $\|\eta\|_2 = 1$, $\mathcal{V}^-$ and $\mathcal{V}^+$ are functions of $P_{\eta^\perp} y$ only, which is independent of $\eta^T y$.

Corollary

The distribution of $\eta^T y$ conditioned on $\{Ay \leq b, P_{\eta^\perp} y\}$ is a (univariate) Gaussian truncated to fall between $\mathcal{V}^-(P_{\eta^\perp} y)$ and $\mathcal{V}^+(P_{\eta^\perp} y)$, *i.e.*

$$\eta^T y \mid \{Ay \leq b, P_{\eta^\perp} y\} \sim TN(\eta^T \mu, \sigma^2 \|\eta\|^2, \mathcal{V}^-, \mathcal{V}^+)$$

$TN(\mu, \sigma, a, b)$ is the normal distribution truncated to lie between a and b .

Theorem

Let $F(x; \mu, \sigma^2, a, b)$ denote the CDF of $TN(\mu, \sigma, a, b)$.
Then $F(\eta^T y; \eta^T \mu, \sigma^2 \|\eta\|^2, \mathcal{V}^-, \mathcal{V}^+)$ is a pivotal quantity

$$F(\eta^T y; \eta^T \mu, \sigma^2 \|\eta\|^2, \mathcal{V}^-, \mathcal{V}^+) \sim \text{Unif}(0, 1).$$

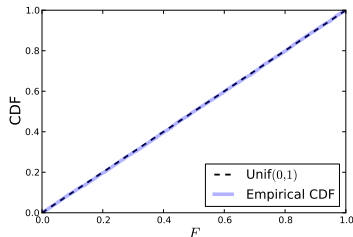
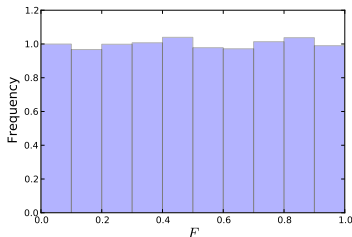


Figure: Pivot is uniform.

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Testing regression coefficients

Coefficients of selected variables are adaptive linear functions

Recall, $\beta^{\hat{M}} = X_{\hat{M}}^{\dagger} \mu$, and $\hat{\beta}_{\hat{M}} = X_{\hat{M}}^{\dagger} y$. By choosing $\eta_j = X_{\hat{M}}^{\dagger T} e_j$, we have $\eta_j^T y = \hat{\beta}_j^{\hat{M}}$.

Theorem

Let $H_0 : \beta_j^{\hat{M}} = \beta_j$. The test that rejects if

$$F(\hat{\beta}_j^{\hat{M}}; \beta_j; \sigma^2 \|\eta\|^2, \mathcal{V}^-, \mathcal{V}^+) \notin \left(\frac{\alpha}{2}, 1 - \frac{\alpha}{2} \right)$$

is a α -level selective test of H_0 .

Other parameter targets

The same result holds exactly for the OLS parameter β^* and β^d .

Algorithm 2 Hypothesis test for selected variables

- 1: **Input:** Design matrix X , response y , model size k .
 - 2: Use variable selection method (marginal screening or Lasso) to select a subset of variables $\hat{M}$.
 - 3: Test $H_0 : \beta_j^{\hat{M}} = \beta_j$.
 - 4: Let $A = A(\hat{M}, \hat{s})$ and $b = b(\hat{M}, \hat{s})$. Let $\eta_j = (X_{\hat{M}}^T)^\dagger e_j$.
 - 5: Compute $F(\hat{\beta}_j^{\hat{M}}; \beta_j, \sigma^2 \|\eta_j\|^2, \mathcal{V}^-, \mathcal{V}^+)$.
 - 6: **Output:** Reject if $F(\hat{\beta}_j^{\hat{M}}; \beta_j; \sigma^2 \|\eta\|^2, \mathcal{V}^-, \mathcal{V}^+) \notin (\frac{\alpha}{2}, 1 - \frac{\alpha}{2})$.
-

Confidence Intervals

Confidence interval C_j is all β_j 's, where a test of $H_0 : \beta_j^{\hat{M}} = \beta_j$ fails to reject at level α .

$$C_j = \{\beta_j : \frac{\alpha}{2} \leq F(\hat{\beta}_j^{\hat{M}}; \beta_j, \sigma^2 \|\eta_j\|^2, \mathcal{V}^-, \mathcal{V}^+) \leq 1 - \frac{\alpha}{2}\}$$

Interval $[L_j, U_j]$ is found by univariate root-finding on a monotone function. Solve

$$F(\hat{\beta}_j^{\hat{M}}; L_j, \sigma^2 \|\eta_j\|^2, \mathcal{V}^-, \mathcal{V}^+) = \frac{\alpha}{2}$$

$$F(\hat{\beta}_j^{\hat{M}}; U_j, \sigma^2 \|\eta_j\|^2, \mathcal{V}^-, \mathcal{V}^+) = 1 - \alpha$$

Lemma (Valid selective confidence intervals)

These are valid selective intervals,

$$\Pr\left(\beta_j^{\hat{M}} \in C_j\right) = 1 - \alpha.$$

False coverage rate (FCR) is the confidence interval analog of FDR. FCR is also controlled.

$$\mathbb{E}\left[\frac{\text{Number not covering and selected}}{\text{Number selected}}\right] \leq \alpha$$

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Comparison to data splitting

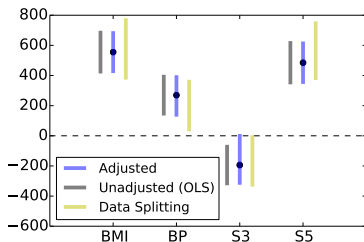


Figure: Diabetes dataset. Blue line is our selective intervals, gray line is z-intervals which ignore selection, and yellow line is the intervals computed using data splitting.

- Variable S3 is no longer significant after adjusting for model selection.
- Our selective intervals are approximately the same as the z-intervals for significant variables. Data splitting widens the intervals by $\sqrt{2}$. Recent follow-up work shows that data splitting is inadmissible and a variant of Condition on Selection dominates data splitting (Fithian, Sun, Taylor).

Non-Gaussian noise and estimated σ^2

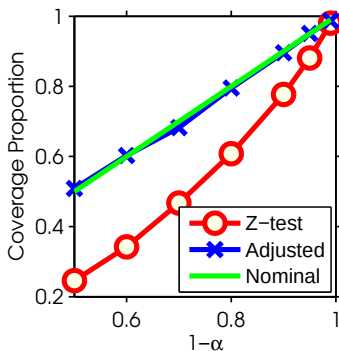


Figure: Plot of nominal coverage vs the actual coverage proportion for diabetes dataset. Simulation is done by using 2000 iterations of residual bootstrap with estimated $\hat{\sigma}^2$. The adjusted intervals always cover at the nominal level, whereas the z-test is always below.

Selective Intervals for sparse β^0 in $p > n$

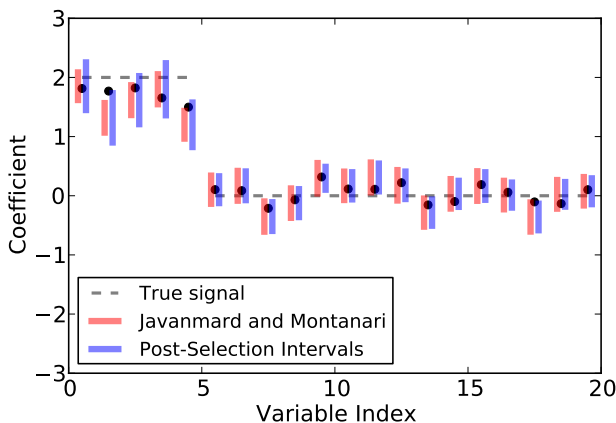


Figure: $(n, p, s) = (25, 50, 5)$ with only the first 20 coefficients being plotted. Data is generated from $y = X\beta^0 + \epsilon$ with a SNR of 2. The Javanmard-Montanari debiased intervals do not guarantee selective coverage.

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Extensions

- Testing the goodness of fit of the selected model,
 $H_0 : (I - P_{\hat{M}})\mu = 0$.
- Pathwise algorithm for stopping Lasso that controls FWER.
- Non-affine regions, only need to intersect a ray with the region to design exact conditional tests.
- Selective t -tests that do not require σ^2 to be known.
Non-Gaussian noise can be handled via CLT (Tian & Taylor 2015).

Easily generalizes to other model selection procedures!

- Orthogonal matching pursuit/ forward stepwise regression, non-negative least squares, and isotonic regression
- LARS (Taylor et al. 2014) and elastic net, and other regularizers.
- AIC, BIC, Models selected via hold-out set or CV.

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Takeaways

- Valid selective inference can be done via the Condition on Selection framework. This results in tests/confidence intervals with exact type 1 error control.
- The selective confidence intervals satisfy

$$\Pr \left(\beta(\hat{M}(y))_j \in C_j \right) = 1 - \alpha.$$

- Test for whether the lasso/marginal screening have found all relevant variables.
- Framework is applicable to many model selection procedures including marginal screening, lasso, forward stepwise, non-negative least squares, AIC, and BIC.

Acknowledgments

Thanks to Trevor Hastie and other members of the Hastie, Tibshirani and Taylor group for feedback.

References:

- ① Jason D. Lee and Jonathan Taylor, *Exact statistical inference after marginal screening*.
- ② Jason D. Lee, Dennis L Sun, Yuekai Sun, and Jonathan Taylor, *Exact post-selection inference with the Lasso*.
- ③ Jason D. Lee, Yuekai Sun, and Jonathan Taylor, *Evaluating the Statistical Significance of Biclusters and other Combinatorial Structures*.

Papers available at <http://stanford.edu/~jdl17/>

Thanks for Listening!

Lasso Selective Intervals

Solve Lasso at some λ , and construct confidence intervals using previous algorithm.

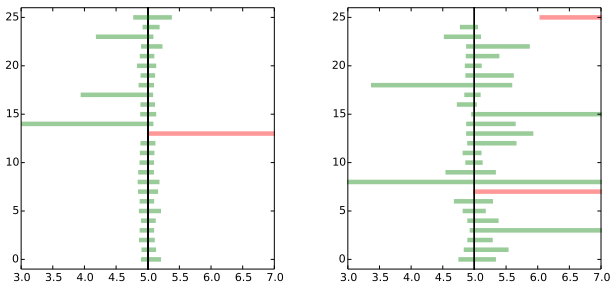


Figure: 90% confidence intervals for $\beta_1^{\hat{M}}$ for two different settings $(n, p) = (100, 50)$ and $(n, p) = (100, 200)$, over 25 simulated data sets. The truth β^0 has five non-zero coefficients, all set to 5.0, and the noise variance is 0.25. A green bar means the confidence interval covers the true value while a red bar means otherwise.

Coarsest selection event

Recall that a subset/sign pair (S, s) is in bijection with a selection event. We only need to condition on the selection for the variables S , which determines η . Selection event for only variables S :

$$\begin{aligned}\{y : \hat{M}(y) = M\} &= \bigcup_{s \in \{-1, 1\}^{|\hat{M}|}} \{y : (\hat{M}(y), s(y)) = (M, s)\} \\ &= \bigcup_{s \in \{-1, 1\}^{|\hat{M}|}} \{y : A(M, s)y \leq b(M, s)\}\end{aligned}$$

- Condition on the coarsest partition where η is still measurable.
- The set is a union of linear constraints. Pivotal quantity, hypothesis tests, and intervals are valid for union of linear constraints.
- Results in shorter confidence intervals, at the cost of more computation.

Coarsest Selection event

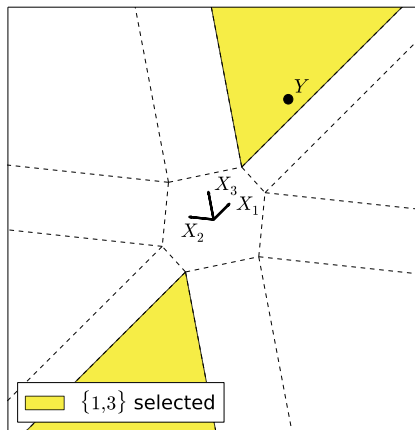


Figure: Coarsest selection event corresponds to the union of the two yellow regions. Before, we conditioned on one of the yellow wedges.

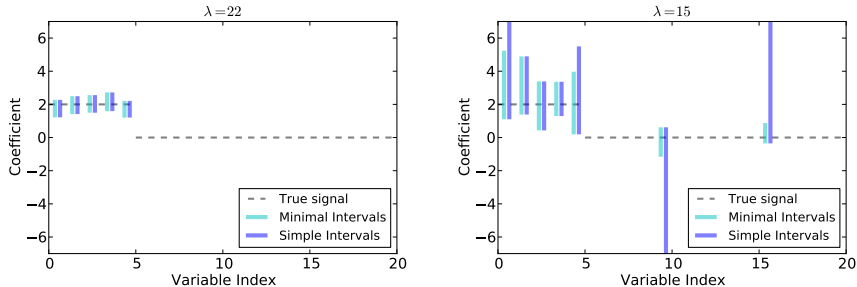


Figure: Light blue intervals are using the coarsest selection event or union of regions and dark blue are using the selection event that is one region. The simulated data featured $n = 25$, $p = 50$, and 5 true non-zero coefficients; only the first 20 coefficients are shown. The simple intervals are as good as the minimal intervals on the left plot; the advantage of the minimal intervals is realized when the estimate is unstable and the simple intervals are very long, as in the right plot.

We would like to test

$$H_0 : \beta_{-\hat{S}}^0 = 0.$$

This means that all the true signal variables have been found,
 $\text{support}(\beta^0) \subset \hat{S}$.

We can test this by checking whether the unselected variables help explain the residual, or $H_0 : \|(I - P_{\hat{S}})\mu\|_{\infty} = 0$.

Testing goodness-of-fit

Letting $j^\star := \operatorname{argmax}_j |e_j^T (I - P_{\hat{S}})y|$ and $s_j := \operatorname{sign}(e_j^T (I - P_{\hat{S}})y)$, we set

$$\eta_{j^\star} = s_{j^\star} (I - P_{\hat{S}})e_{j^\star},$$

and test $H_0 : \eta_{j^\star}^T \mu = 0$. This is a linear function of y .

Corollary

Let $H_0 : \|(I - P_{\hat{S}})\mu\|_\infty = 0$. Then, the test which rejects when

$$\left\{ F_{0, \sigma^2 \|\eta_{j^\star}^*\|^2}^{[\mathcal{V}^-, \mathcal{V}^+]}(\eta_{j^\star}^T y) > 1 - \alpha \right\}$$

is level α ,

$$\mathbb{P} \left(F_{0, \sigma^2 \|\eta_{j^\star}^*\|^2}^{[\mathcal{V}^-, \mathcal{V}^+]}(\eta_{j^\star}^T y) > 1 - \alpha \mid H_0 \right) = \alpha.$$

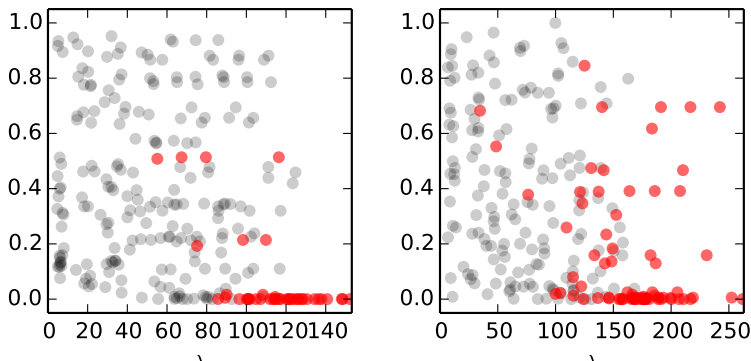


Figure: P-values for $H_{0,\lambda}$ at various λ values for a small ($n = 100$, $p = 50$) and a large ($n = 100$, $p = 200$) uncorrelated Gaussian design, computed over 50 simulated data sets. The true model has three non-zero coefficients, all set to 1.0, and the noise variance is 2.0. We see the p-values are $\text{Unif}(0, 1)$ when the selected model includes the truly relevant predictors (black dots) and are stochastically smaller than $\text{Unif}(0, 1)$ when the selected model omits a relevant predictor (red dots).

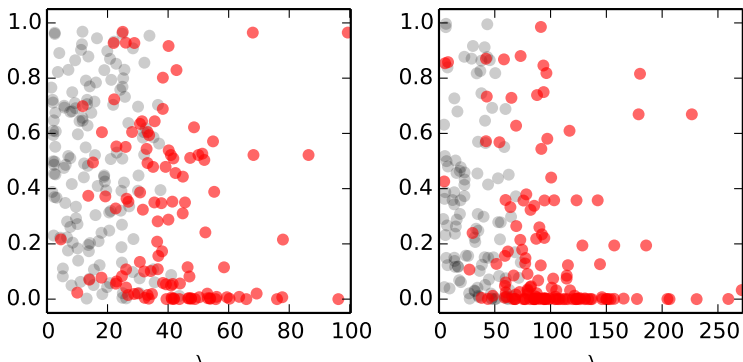


Figure: P-values for $H_{0,\lambda}$ at various λ values for a small ($n = 100, p = 50$) and a large ($n = 100, p = 200$) *correlated* ($\rho = 0.7$) Gaussian design, computed over 50 simulated data sets. The true model has three non-zero coefficients, all set to 1.0, and the noise variance is 2.0. Since the predictors are correlated, the relevant predictors are not always selected first. However, the p-values remain uniformly distributed when $H_{0,\lambda}$ is true and stochastically smaller than $\text{Unif}(0, 1)$ otherwise.

- 1 **False discovery rate control in linear regression.** Selective inference with the Benjamini-Yekutieli procedure ensures FDR control in the $n > p$ regime. By combining with recent work on debiased estimators β^d , we can ensure FDR control in sparse high-dimensional linear regression.
- 2 **Scan statistics and approximate scan statistics.** We work out an exact test based off the scan statistic

$$z^* = \max_{C \in \mathcal{S}} \sum_{i \in C} z_i.$$

Frequently scan statistics are too expensive to compute, and greedy/approximate algorithms are used to approximate the scan statistic. We work out an exact test for these algorithms too, using the Condition on Selection framework.